

Future predictions from current plant collections: Exploiting the novel climates of arboreta to understand tree responses to climate change**Introduction**

Understanding how climate affects species' performances and distributions has been a focus in ecology for centuries (Darwin 1859, Grinnell 1917), and is essential to preparing for climate change. One major effect of increasing global temperatures will be large-scale changes to forests (Reich and Oleksyn 2008, van Mantgem et al. 2009, Harsch et al. 2009), but exactly how tree growth and survival will be affected by climate change is uncertain. Studies to date suggest that temperate and boreal tree responses to warming range from growing faster, slower, or at unchanged rates (Reich and Oleksyn 2008, Ettinger et al. 2011).

Understanding such divergent results requires new approaches that can identify the drivers of these inconsistencies, and previous research on tree growth is inadequate for two major reasons. First, studies almost always examine trees growing in their native geographic ranges. However, climate change will alter climates beyond historic variation: much of the world will experience novel climates by 2100 (Williams et al. 2007). Second, most studies, including the few that have investigated trees transplanted outside their native ranges, focus on a single species (Reich and Oleksyn 2008). This has not enabled generalizations across species to be made. For example, is climate change sensitivity primarily determined by traits (*e.g.* wood density, phenology), phylogenetics (*e.g.* family or genus, which can also constrain traits, Kingsolver et al. 2011), or the amount of climatic change experienced?

Functional trait analysis is a valuable approach for understanding past inconsistencies in climate change responses, because comparisons can be made across species, ecosystems, and timescales (Polly et al. 2011). Tradeoffs between different aspects of climate response, such as between tolerance to high versus low temperature, may determine an individual's overall performance in a given climate, and these responses are mediated by functional traits (Westoby and Wright 2006, Angert et al. 2011). In trees, changes in climate will affect photosynthesis, respiration, and tissue composition (Hughes 2000); whether shifts in these processes benefit or harm an individual's performance may be predictable based on particular traits. For example, wood density can predict drought sensitivity (Westoby and Wright 2006) and the timing of growth onset (phenology) is an important predictor of frost damage (Hanninen 2006).

Species' evolutionary histories may also affect responses to climate change. Traits governing climate responses can be phylogenetically conserved, suggesting that trait analysis should be conducted in a phylogenetic context (Kingsolver et al. 2011). If climate sensitivity shows high phylogenetic conservatism, this may facilitate forecasting efforts, as it would suggest that climate change responses of a few taxa can be generalized to closely related species. Few studies have explicitly examined phylogenetic patterning in climate change responses (Kingsolver et al. 2011), and we know of none that focus on tree growth responses specifically.

In addition to traits and their phylogenetic context, forecasting climate change responses requires incorporating variation in climatic shifts across the globe. For example, temperatures have increased more in polar and temperate than tropical areas (IPCC 2007). Regions experiencing greater magnitudes of change may correspond to larger biological responses. The extent to which climate controls performance is uncertain, however, since biotic interactions and other nonclimatic factors also affect performance (*e.g.* Ettinger et al. 2011, Ettinger and HilleRisLambers 2013).

I propose to study how traits, phylogenetics, and variation in climatic change affect interspecific differences in climate sensitivity, using annual tree growth at Arnold Arboretum (Fig. 1). The Arboretum provides a unique opportunity to address limitations of previous studies to advance climate change research. Its unrivaled collection of diverse woody species growing

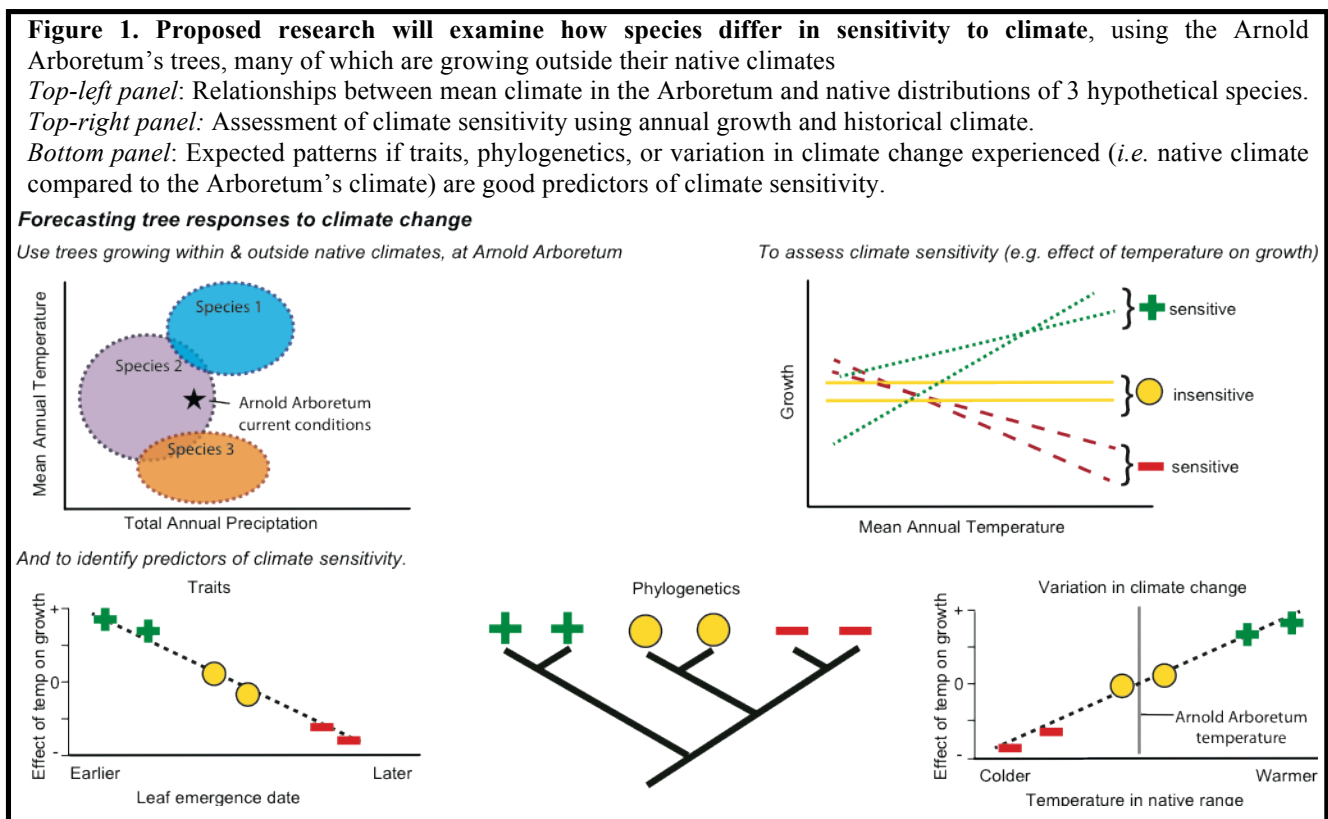
together enables traits to be studied in a phylogenetic context. Furthermore, many species in the Arboretum's collections are nonnative and have been exposed to a "novel" climate that may resemble future conditions in their native distributions (Fig. 2).

Proposed Research Questions & Approach

I propose to combine techniques from climatology (tree ring studies), ecology (traits analysis), and evolution (phylogenetic relationships). Using historic climate data with the Arboretum's extensive living collections, I will:

- 1) *Assess climate sensitivity*: Across diverse species, how has tree growth responded to past climate variation?
- 2) *Forecast future responses to climate change*: Given past growth-climate relationships, how will future climate change affect tree growth?
- 3) *Identify predictors of climate change sensitivity*: Are there characteristics (*i.e.* traits such as wood density, leaf venation and leaf size; phylogenetic relatedness; or variation in climate change experienced, relative to climate in the native range) that can be used to forecast the direction and/or magnitude of species' responses to climate (Fig. 1)?

I plan to conduct this work primarily in the lab of Dr. Elizabeth Wolkovich at Arnold Arboretum.



Methods

To quantify climatic influences on growth, I will compare historic climate data to annual growth (attained by coring trees) of 19 tree genera across 11 families. I will use mixed effects models to identify which climate variables, if any, best explain tree growth and assess climate sensitivity of each species (Fig. 1). I will use forecasted changes of these climate variables, as well as a global vegetation model, to understand how future growth may be affected. Next, I will compare climate sensitivity across species to examine potential predictors of sensitivity (Fig. 1).

Tree growth data— I will core trees at Arnold Arboretum and use standard dendrochronological techniques to measure and date ring widths to calendar years (Stokes and Ailene K. Ettinger

Smiley 1968, Salzer et al. 2009, Chen et al. 2010, Ettinger et al. 2011). For a subset of species, I will also assemble growth data from native distributions, using cores from the International Tree Ring Data Bank and other sources (e.g. Ettinger et al. 2011), and by collecting cores from species in Boston and at Harvard Forest. This will allow me to compare climate sensitivity within versus outside species' realized climate niches, and in a managed landscape versus unmanaged areas.

Climate data and forecasts—To quantify growth-climate relationships, I will use National Climate Data Center¹ data to calculate seasonal and annual climate variables (e.g. mean annual, growing season, and dormant season temperature; total precipitation; growing degree days). I will use Climate Research Unit data² to quantify species' realized climate niches, and downscaled projections for forecasted changes in climate (Maurer et al. 2007).

Trait data—In focal individuals, I will measure wood density, specific leaf area, and vein structure. I will use growth chambers to test cold tolerances and phenological cues, and collaborate with other researchers to acquire additional phenological data (e.g. leaf budburst date, (Primack et al. 2004, Miller-Rushing et al. 2006, Del Tredici 2013)). Other traits can be attained from floras (e.g. Haines 2011), scientific literature, and online databases.^{3,4}

Phylogenetic data—I will assemble a phylogeny for focal species first using Phylomatic⁵ with the APG III backbone. To achieve finer resolution, I will use genetic data to infer a maximum likelihood phylogeny with available DNA sequence data from GenBank (Davies et al. 2013, Wolkovich et al. 2013). The resulting phylogeny will be used to estimate branch lengths (i.e. “relatedness”) between focal taxa.

Realized climate niche data—I will use presence-absence distribution data⁶ and global temperature and precipitation data² to estimate the temperature and precipitation estimates that make up each species' realized climate niche (i.e. climate in its native range). I will compare realized climate niches to Boston climate to quantify the magnitude of climate change experienced by individuals at the Arboretum, relative to native climates.

Statistical analyses—To assess climate sensitivity, I will fit mixed effects models with tree growth as the response and climate variables (individually and in combination) as potential explanatory variables, to identify which climate variables, if any, best explain tree growth. Individual tree and year will be included as random effects to account for non-independence (Crawley 2007). To forecast future responses to climate change, I will use these relationships, combined with forecasts for climate changes in Massachusetts, to understand how growth is likely to change. I will also compare results to forecasted growth using the Ecosystem Demography model, a global vegetation model that employs climate, light, and other inputs to predict tree growth (Medvigy et al. 2009, Medvigy and Moorcroft 2012). To identify predictors of climate change sensitivity, I will test the ability of traits, phylogenetic relatedness, and/or variation in climate change experienced to predict climate sensitivity of tree growth. All analyses will be conducted in the open-source program R (R Core Team 2013, Bunn 2010).

Safely Utilizing the Collections—Coring trees is nondestructive (van Mantgem and Stephenson 2004, Wunder et al. 2011). I will sterilize increment borers between individuals to reduce disease transmission risk. Prior to coring any trees, I will get approval from Arboretum staff.

Laboratory and Equipment Requirements—At least two Hagloff 5.15 mm-diameter 16” (or greater) increment borers, a hand-held sander, balances, a drying oven, a dissecting microscope or a high resolution scanner attached to a computer with the ring identification program WinDendro and a computer with ArcGIS software.

Anticipated Results and Implications

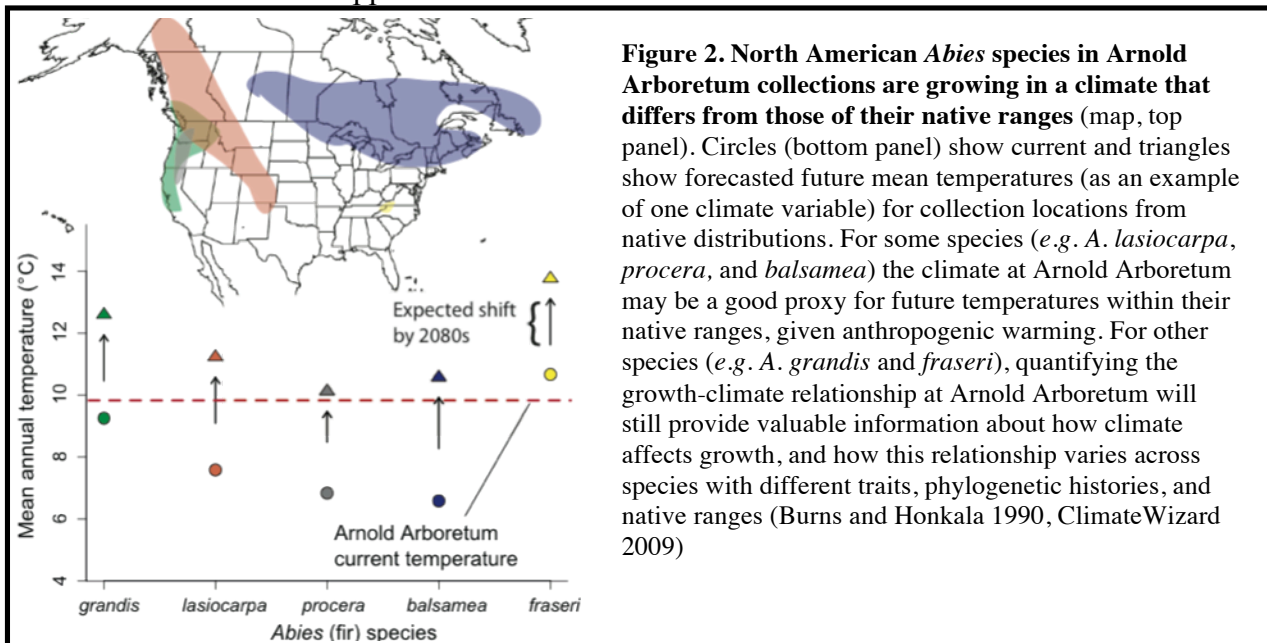
A major challenge for ecology is to be more predictive (Evans et al. 2013). This research accomplishes critical insights by addressing the role of climate in controlling performance across

diverse species. It also provides a novel approach and opportunity to test existing forecasting frameworks (Medvigy et al. 2009, Buckley and Kingsolver 2012).

My research will identify predictors for climate sensitivity that may be relevant worldwide, and can inform future studies of mechanisms behind climate responses. For example, for *Abies* (fir), my study would provide growth-climate information for species with native distributions in Asia, Europe, and North America (Fig. 2). I expect that some traits, such as earlier leaf emergence, will be associated with greater temperature sensitivity, and that others, such as higher wood density, will be associated with reduced sensitivity (Bouriaud et al. 2004). I also expect highest sensitivity in species whose native climate niches do not encompass the Arboretum's climate. I expect growth of these climate-sensitive species to be positively influenced by temperature, if native ranges are warmer than Boston (e.g. *A. fraseri*, Fig. 2).

My findings can also be used to improve forecasts of future biological impacts of climate change. Tree growth is correlated with other aspects of performance, such as reproductive success and mortality (Despland and Houle 1997, Wyckoff and Clark 2000). Thus, my results may be indicative of climate change “winners” and “losers,” at Arnold Arboretum and elsewhere. Moreover, I will use my results to test forecasts of tree growth based on the Ecosystem Demography model, a global vegetation model that uses climate, radiation, and other inputs to predict tree growth (Medvigy et al. 2009, Medvigy and Moorcroft 2012).

The proposed research offers many opportunities for collaboration and outreach. This approach could become a widespread tool for forecasting. I will therefore give talks at other arboreta to share methods and results; I hope to sample their collections, as well, to expand the climates and species over which inferences can be made. Furthermore, I will use collected tree cores and other materials to develop climate change curricula, for use in the Arboretum's educational programs and Boston Public Schools. I plan to publish at least two papers per year in *Arnoldia* and other scientific journals, and will present findings at professional meetings (e.g. Ecological Society of America, American Geophysical Union), to natural resource managers, and other audiences for whom there are relevant applications.



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¹ <http://www.ncdc.noaa.gov/>

² <http://www.cru.uea.ac.uk/data>

³ <http://nora.nerc.ac.uk/9535/>

⁴ <http://plants.usda.gov>

⁵ <http://phylodiversity.net/phyloomatic/>

⁶ <http://esp.cr.usgs.gov/data/little/>